CULTIVATION GUIDE | BACKGROUND

Sri Lankan Paddy: Seasons, Water, Growth Stages and Varieties

The local facts the other guides depend on.

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Teaching edition for the STEMLink AI Engineer Bootcamp. Content restructured from AgriVibe knowledge base AV-KB-001 Rev 1.1, which grounds its agronomy in Department of Agriculture and RRDI guidance for Sri Lanka. Not for use with real farmers until the verification items in AV-POL-03 are closed.

1. Growing seasons

Paddy is grown in two seasons, each tied to a monsoon. Maha is the main season, running roughly September to March under the North-East monsoon and producing around two thirds of national rice output. Yala is the minor season, running roughly May to August under the South-West monsoon, generally with less water available. A season is defined by when the crop is sown and harvested within those periods.

2. Water and irrigation

Rice is grown as a rainfed crop, or with supplementary or fuller irrigation from tanks and anicuts through distributary channels. Rainfall in rice-growing areas normally meets at least part of the crop's requirement, so in strict terms there is no fully irrigated rice cultivation in the country.

Rice is a semi-aquatic plant and does not actually require standing water to succeed. It is grown with standing water mainly because water supply is uncertain and because flooding suppresses weeds. Bunds are maintained to retain water and prevent leakage from the field. A direct-sown crop is generally less prone to drought than a transplanted crop.

> **IMPLICATION**
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> Whether a field is irrigated or rainfed changes both what the farmer can do about water and which fertiliser schedule applies (AV-CG-06). Where this is unknown, it must not be assumed. Check the field registry (AV-FR-00).

3. Crop age classes and growth stages

Departmental recommendations are organised around varietal age classes of roughly three months, three and a half months and four months from establishment. Age determines both the position in the fertiliser schedule and the growth stage reached.

Table 1. Approximate crop age to growth stage, for interpretation

0 to 2 weeks. Stage: Establishment; NDVI expectation: Low; much water and soil visible.
2 to 6 weeks. Stage: Tillering; NDVI expectation: Rising.
6 to 9 weeks. Stage: Panicle initiation to booting; NDVI expectation: Near peak.
9 to 12 weeks. Stage: Flowering to grain filling; NDVI expectation: High, then beginning to fall.
Final weeks. Stage: Ripening; NDVI expectation: Falling naturally toward harvest.

4. Varieties

Sri Lankan improved rice varieties are commonly identified by prefixes reflecting the research station that developed them: Bg for Bathalagoda, Bw for Bombuwala, At for Ambalantota and Ld for Labuduwa. Varieties differ in tolerance to stresses including iron toxicity. A variety name mentioned by a farmer should be

recognised as a variety identifier; the field record states its age class, and AV-CG-07 gives the published details of the varieties in AgriVibe fields.

5. Institutions

The Department of Agriculture is headquartered at Peradeniya. The Rice Research and Development Institute at Batalagoda in the Kurunegala district is the national centre for rice research, with regional stations including Bombuwala, Ambalantota, Labuduwa, Bentota, Sammanthurei, Murunkan and Paranthan. The National Agriculture Information and Communication Centre operates the 1920 advisory service and the Krushi Advisor mobile application.